Two clocks, not one: measured pressure–temperature decoupling, its operator origin, and the structural blind spot of single-diffusivity closure

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Abstract

Pressure and temperature in a fluid are driven by the same source yet evolve under different operators: temperature obeys a local, parabolic advection–diffusion equation, while pressure in the incompressible limit obeys a global elliptic constraint — the finite-Mach hyperbolic acoustic adjustment degenerating into it as $M \to 0$. This paper assembles the direct observational and computational case that this operator split is measurable as *two clocks* running in one flow, and that single-eddy-diffusivity ("K-theory") closure is structurally blind to it. **Temporally**: in NEON eddy-covariance data (WREF, 30-min, winter and summer months) temperature is a clean diurnal (24 h) signal while barometric pressure is synoptic-plus-tidal, and the bulk daily P–T correlation is regime-dependent but never gas-law-locked ($r \approx +0.07$ in winter, -0.49 in the summer thermal-low regime, against a constant-density lock slope of +0.33 kPa/K); in ASOS 1-minute data at eight stations spanning 24.6°–47.4°N the 12-h pressure tide amplitude falls monotonically $1.10 \to 0.34$ hPa toward the pole (daily-block-bootstrap 95% CIs; trend correlation 0.98 with the Haurwitz/Dai–Wang migrating-tide climatology $1.16 \cos^3 \varphi$, station totals 0.9–1.4× the migrating-only formula as non-migrating components require) while the 24-h temperature amplitude stays local — pressure carries a planetary-scale resonance where temperature carries the sun overhead; band-resolved coherence completes the split: 0.8–1.0 at the solar lines (the shared astronomical clock, above a phase-randomized surrogate 95% level at all eight stations) collapsing to the surrogate noise floor in the off-line sub-diurnal band. **Spatially**: in a $Ra = 10^6$ Rayleigh–Bénard DNS the two fields share one integral scale (ratio 1.06×) yet the pressure Taylor microscale is 2.40× the buoyancy's and the high-wavenumber pressure spectrum sits $\sim 10^4\times$ below buoyancy — the $1/k^2$ footprint of the inverse Laplacian; sweeping $Ra = 10^6 \to 10^8$ *widens* the microscale ratio to 4.5× while the shared integral scale stays order one, so the split strengthens with turbulence intensity; correspondingly, a localized boundary bump projects 79% of its elliptic pressure response beyond one bump-width while the matched parabolic response stays confined. The **closure-relevant consequence** is measured directly: across Monin–Obukhov stability classes the momentum-to-heat transport-efficiency ratio swings 4.3× ($0.77 \to 3.28$) in the winter month and 6.0× in the summer month, structurally outside any fixed-turbulent-Prandtl single-diffusivity closure. Three **controlled demonstrations** isolate the mechanism: (i) a compressible pressure field relaxes to the elliptic Poisson field with $t_{90} \approx 2.3\, L/c$, all sound speeds collapsing onto one curve in ct/L; (ii) in a fully nonlinear compressible box the dilatational (fast-clock) energy fraction scales as

M^2 and the acoustic-averaged pressure converges to the elliptic field (correlation ≈ 1, residual $\sim M^2$); (iii) the projection method is exhibited as the two-clocks structure made literal — one Leray solve reduces the local drift's divergence from 2.7×10^{-2} to machine zero (2×10^{-15}), whereas phase-randomized surrogates with the *exactly* matched power spectrum leave it at 3.7×10^{-2} [3.5–3.8×10^{-2}, 95% over $n = 100$ surrogates] — no better than applying no correction at all (2.7×10^{-2}) — with pointwise correlation to the true correction of median $|r| = 0.07$ (95th percentile 0.20; the seed ensemble replaces a single-realization anecdote) — a structural ceiling of spectrum-matching stochastic closures. Finally, the same decomposition applied to **real winds** (NCEP/NCAR reanalysis, spherical-harmonic Helmholtz split) yields $KE_{div}/KE_{rot} = 4.0\%$, 0.7%, 1.0% at 850/500/250 hPa with the minimum at the classical level of non-divergence — reproduced in the independent NCEP–DOE Reanalysis 2 (4.5%, 0.8%, 1.1%, same ordering, same 500-hPa minimum; $\leq 15\%$ relative differences), so the ratio is not an assimilation-system artifact — and daily averaging scrubs the fast clock exactly as acoustic averaging does in the box. Scope is explicit: the demonstrations are two-dimensional and periodic; no 3-D regularity claim and no forecast-skill claim is made; reanalysis is a model-assimilated confirmation at $l \leq 35$. A companion paper derives, audits, and benchmarks the closure repair; this paper establishes the phenomenon it repairs.

1. INTRODUCTION

1.1 One source, two responses

Turn on the sun over a fluid layer and two of its most basic state variables respond in ways that could hardly be more different. Temperature responds *locally*: it is advected, it diffuses, it remembers the surface that heated it, and its anomalies are torn into filaments by whatever strain the flow supplies. Pressure responds *globally*: in the incompressible limit it is not a prognostic field at all but the multiplier of the divergence-free constraint, resolved instantaneously over the whole domain and its boundaries; at finite compressibility that instantaneous adjustment is replaced by acoustic waves running at a speed far faster than the flow. The two fields are coupled — through buoyancy, through the equation of state, through the momentum balance — and driven by the same heating; yet they carry different

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histories, different geometries, and different characteristic times. We call this the *two clocks* structure: a slow, local, memory-bearing clock (temperature and other transported scalars) and a fast, global, constraint-like clock (pressure).

The distinction is elementary at the level of the equations. It stops being elementary the moment one builds a turbulence closure, a subgrid parameterization, or a statistical forecast model, because the workhorse of all three — a single scalar eddy diffusivity acting on resolved gradients (Boussinesq 1877; Smagorinsky 1963), optionally with a fixed turbulent Prandtl number to relate momentum and heat — assumes *one* clock. Everything transported is assumed stirred by the same eddies at the same rate with no memory and no action at a distance. The premise of this paper is that the two-clocks structure is not a metaphor but a measurable property of real data and simple flows, with a quantifiable failure mode for one-clock closures.

1.2 What this paper does

We proceed from measurement to mechanism to consequence:

1. **Measure the temporal split in the field** (§3). NEON eddy-covariance data and multi-station 1-minute ASOS barometry/thermometry show temperature locked to the local diurnal cycle while pressure is dominated by synoptic variability plus the semidiurnal solar tide — a *global* resonance whose amplitude follows latitude, not local heating. Bulk daily P–T coupling at a fixed site is regime-dependent and never gas-law-locked (winter $r \approx +0.07$, summer $r \approx -0.49$).

2. **Measure the spatial split on one grid** (§4). In a Rayleigh–Bénard DNS, pressure and buoyancy share the same integral (roll) scale but differ 2.40× in Taylor microscale and $\sim 10^4\times$ in high-k spectral power — the direct footprint of the inverse Laplacian's $1/k^2$; the microscale ratio *widens* to 4.5× as Ra is swept $10^6 \to 10^8$, and a Green's-function experiment makes the same point with a single localized source.

3. **Measure the closure-relevant consequence** (§5). Binned by Monin–Obukhov stability, the ratio of momentum to heat transport efficiency swings 4.3× where a fixed-Pr_t closure requires it constant.

4. **Demonstrate the mechanism in controlled solvers** (§6). A linear acoustics solver shows the hyperbolic→elliptic crossover with $t_{90} \approx 2.3\,L/c$; a nonlinear compressible solver shows both clocks coexisting with the fast clock's energy $\sim M^2$; a Boussinesq solver shows that the projection method *is* the two-clocks synthesis, and that a spectrum-matched stochastic surrogate cannot reproduce the elliptic correction (median $|r| = 0.07$, p95 0.20, $n = 100$; divergence unrepaired) — the structural ceiling of SPDE-style closures.

5. **Confirm on real winds** (§7). A spherical-harmonic Helmholtz split of NCEP/NCAR reanalysis puts $\lesssim 4\%$ of kinetic energy in the divergent (fast) component, with the vertical structure pinned to the classical level of non-divergence and time-averaging acting as the same low-pass filter as acoustic averaging in the box.

Section 8 states scope and kill conditions for each claim. A companion manuscript (Saifelden 2026, hereafter **P1**) takes the diagnosis to prescription: it audits eddy-diffusivity closure operator-by-operator against the Mori–Zwanzig generalized Langevin equation and benchmarks a projected, fluctuation–dissipation-consistent repair in 2-D and 3-D. This paper is the evidence layer: every number above is regenerable from committed code (Appendix A), and none of it depends on P1's closure machinery.

1.3 Relation to existing understanding

None of the ingredients is exotic. That low-Mach-number flow relaxes to an elliptically-constrained state is classical singular-limit theory (Klainerman & Majda 1981); atmospheric tides are a mature subject (Chapman & Lindzen 1970; Dai & Wang 1999); the rotational dominance of large-scale winds and the level of non-divergence are textbook synoptic dynamics (Holton 2004); projection methods are standard numerics (Chorin 1968; Leray 1934); and stochastic parameterization is an active program (Hasselmann 1976; Berner et al. 2017). The contribution here is the *joint, quantitative* case built for the closure question: the same operator split measured in field data, in a DNS, in controlled compressible and incompressible solvers, and in reanalysis, each with a falsifiable number attached, and each mapped onto the specific term of the closure problem it breaks. In particular the spectrum-matching ceiling of §6.3 — energy-consistent noise reproducing the fast clock's *spectrum* while leaving its *geometry* unreconstructed (ensemble-median correlation 0.07, the divergence constraint unrepaired) — is, to our knowledge, not stated quantitatively elsewhere, and it is the piece that connects the phenomenology to closure structure.

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2. TWO OPERATORS, TWO CLOCKS

2.1 The equations

Let u be velocity, p pressure, θ a transported scalar (temperature, buoyancy, humidity), ν, κ molecular coefficients. The scalar obeys a **parabolic** transport equation,

$$\partial_t \theta + (u \cdot \nabla)\theta = \kappa \nabla^2 \theta + S_\theta, \tag{1}$$

whose Green's function is (in the diffusive limit) a spreading Gaussian of width $\sqrt{4\kappa t}$: *local*, causal, memory-bearing. In an incompressible flow, pressure obeys the **elliptic** Poisson constraint obtained from the divergence of the momentum equation,

$$\nabla^2 p = -\rho \, \nabla \cdot [(u \cdot \nabla) u] + \nabla \cdot f, \tag{2}$$

whose Green's function is $\sim \ln r$ in 2-D ($\sim -1/r$ in 3-D): *global*, instantaneous, boundary-aware. In Fourier space $\hat{p} = -\widehat{f}/k^2$ — division by k^2 is a global low-pass filter, and this single algebraic fact drives every spatial measurement in §4. At finite compressibility pressure is instead **hyperbolic**,

$$\frac{1}{c^2} \, \partial_{tt} p - \nabla^2 p = s(x, t), \tag{3}$$

with sound speed c; as $M = U/c \to 0$, (3) degenerates into (2) — the waves become infinitely fast and vanish into the constraint (Klainerman & Majda 1981). The two descriptions are the same physics at different compressibility, which is exactly what §6.1–6.2 demonstrate numerically.

2.2 The clocks

Three timescales matter for a domain of size L, flow speed U, eddy scale ℓ:

$$\tau_p \sim L/c \;\; \text{(acoustic/elliptic adjustment)}, \qquad \tau_{adv} \sim \ell/U \;\; \text{(advective turnover)}, \qquad \tau_\kappa \sim \ell^2/\kappa \;\; \text{(diffusion)}. \tag{4}$$

For air at laboratory-to-mesoscale L, $\tau_p/\tau_{adv} = M\,(L/\ell)^{-1} \ll 1$; for water ($c \approx 1500$ m/s) smaller still. Pressure equilibrates before the flow moves; scalars integrate their history as the flow moves. A closure that models both with one local diffusivity therefore commits to a specific, testable set of errors:

- **E1 (temporal).** At a fixed site, low-frequency pressure and temperature should decouple: temperature tracks the local source spectrum; pressure tracks global/planetary modes. A one-clock picture in which "P tracks T" through the gas law at fixed density predicts tight daily coupling — so measuring the coupling tests the picture. (§3)

- **E2 (spatial).** On one grid, pressure must lack small-scale power relative to a scalar by the $1/k^4$ spectral factor implied by $\hat{p} \propto \hat{f}/k^2$, even where both share the energy-containing scale. (§4)

- **E3 (closure).** Momentum (shaped by the global pressure field) and heat (shaped by local buoyant plumes) cannot maintain a constant transport-efficiency ratio as the balance of shear and buoyancy changes; a fixed-Pr_t closure says they must. (§5)

- **E4 (crossover).** Making c finite must turn the constraint into waves whose only role, in the slow limit, is to set the adjustment clock: relaxation time $\propto L/c$, final field $\to$ the elliptic solution, fast-clock energy $\to 0$ as M^2. (§6.1–6.2)

- **E5 (structure, not spectrum).** The fast clock's correction to the slow dynamics is a global elliptic solve, not noise: a stochastic surrogate with the correct spectrum but random phases must fail to reproduce it pointwise and must fail its constraint function. (§6.3)

- **E6 (real flow).** In the real atmosphere the slow clock (rotational, balanced) must dominate the kinetic energy, the fast clock (divergent) must be weak and physically structured, and time-averaging must act as a low-pass filter on it. (§7)

Each expectation is falsifiable, and §8.2 lists what observation would have killed — or would still kill — each one.

3. TEMPORAL DECOUPLING MEASURED IN THE FIELD

3.1 NEON eddy covariance: a winter month at one tower, cross-checked in summer

We use the NEON bundled eddy-covariance product (DP4.00200.001) at WREF (Wind River Experimental Forest, WA; 45.82°N, 121.95°W; old-growth conifer canopy, 53 m), January 2020 at 30-min resolution: 1488 intervals, of which 1097 carry valid pressure and 1180 valid temperature. Three facts (Fig. 1):

- **The source–response chain is causal and ordered.** Incoming shortwave leads air temperature by ∼2 h and the sensible heat flux by ∼3 h; daytime flux is upward (buoyant). The temperature clock is driven, locally, by the sun.

- **The two spectra are qualitatively different.** Temperature variance is overwhelmingly diurnal (24 h). Pressure variance is dominated by multi-day synoptic weather plus a distinct **semidiurnal (12 h) atmospheric tide** — twin daily maxima visible in the raw composite.

- **Bulk coupling is near zero.** Pearson $r(P, T) \approx +0.07$ ($R^2 \approx 0.005$) at the daily scale: nowhere near the steep constant-density line a fixed-parcel ideal-gas lock would predict. The mismatch is absorbed by ∼1% density changes — the gas law holds *locally*; what fails is the one-clock reading "bulk P tracks bulk T".

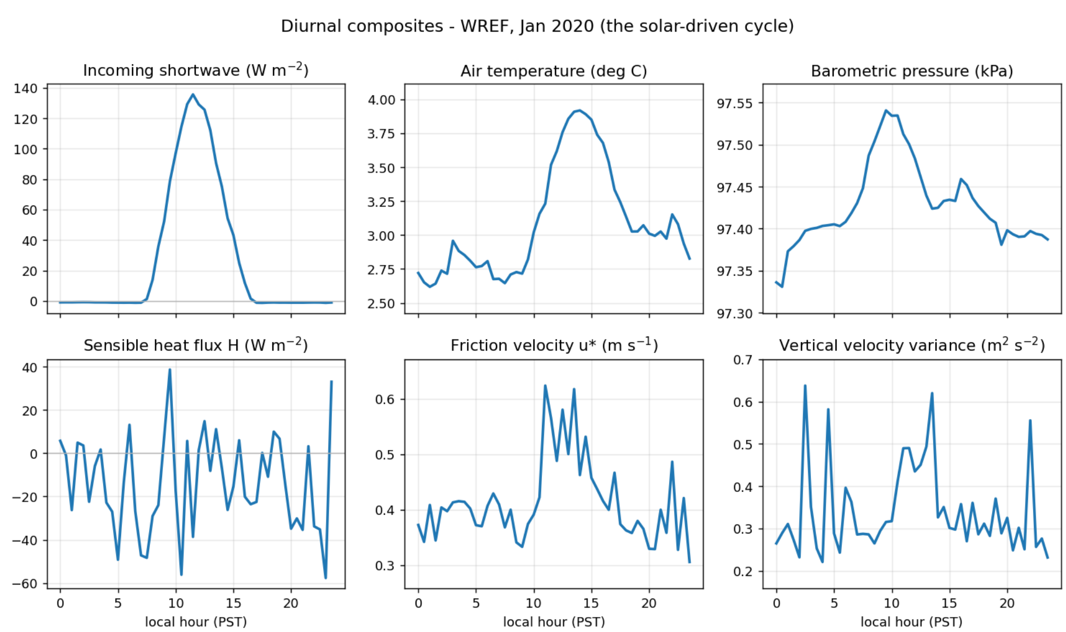

Figure 1. NEON WREF, Jan 2020: diurnal composites of shortwave, temperature, sensible heat flux, and pressure. Temperature is a single clean 24-h signal; pressure shows the twin-peak semidiurnal tide riding on synoptic variability.

Seasonal cross-check (summer month). The same pipeline applied to July 2020 at the same tower (1036/1344 valid P/T intervals) reproduces the fingerprint at the opposite solar extreme: temperature's semidiurnal-to-diurnal power ratio drops to 1.0% (the summer diurnal line is even cleaner; winter 7.3%), pressure keeps a genuine tide line at 11.9 h (8.5% of its diurnal power) with its dominant variability still synoptic (∼168 h), and the bulk daily correlation moves to $r = -0.49$ — the summer thermal-low regime, in which hot afternoons *lower* surface pressure — with observed slope −0.024 kPa/K against the constant-density lock's +0.334 kPa/K. The bulk P–T coupling is set by synoptic regime and season; the operator fingerprint is not (artifact `01b_neon_seasonal.json`).

3.2 ASOS 1-minute: the latitude fingerprint of a global mode

A single tower cannot distinguish "pressure is noisy" from "pressure listens to the whole atmosphere". Eight ASOS stations spanning 23° of latitude, coastal and interior (1-minute data, Q1 2020; Iowa Environmental Mesonet archive; daily moving-block bootstrap, $n = 500$, for the 95% CIs) can:

station	latitude	$S_{1,T}$ (24 h)	$S_{2,P}$ (12 h) [95% CI]	$S_{2,P}/1.16\cos^3\varphi$	daily P–T corr [95% CI]
EYW	24.6°N	1.51 °C	**1.103 [1.013, 1.191] hPa**	1.26	−0.16 [−0.30, −0.03]
MIA	25.8°N	2.73 °C	1.077 [0.979, 1.169] hPa	1.27	−0.05 [−0.18, +0.08]
MSY	30.0°N	2.75 °C	1.018 [0.878, 1.169] hPa	1.35	−0.40 [−0.54, −0.24]
DFW	32.9°N	3.28 °C	0.957 [0.794, 1.141] hPa	1.39	−0.48 [−0.59, −0.34]
PHX	33.4°N	5.06 °C	0.924 [0.861, 1.008] hPa	1.37	−0.20 [−0.29, −0.12]
DSM	41.5°N	2.73 °C	0.628 [0.398, 0.965] hPa	1.29	−0.56 [−0.67, −0.41]
CAR	46.9°N	3.03 °C	0.439 [0.181, 0.879] hPa	1.19	−0.42 [−0.52, −0.30]
SEA	47.4°N	1.79 °C	**0.337 [0.262, 0.529] hPa**	0.94	−0.06 [−0.21, +0.09]

The 24-h temperature amplitude is set by local insolation and does not order with latitude in this winter-season sample (interior PHX largest, marine SEA/EYW smallest). The 12-h pressure tide **falls monotonically from equator to pole across all eight stations**, the classical signature of the global solar semidiurnal tide — a planetary normal-mode response, largest where the forcing projects onto the low-latitude waveguide (Chapman & Lindzen 1970). This is deliberately a **protocol validation against a known planetary signal**, not a discovery claim: the trend correlation with the migrating-tide climatology $A_H = 1.16\cos^3\varphi$ hPa (Haurwitz 1956) is 0.98, and station totals sit 0.9–1.4× A_H, the expected excess once non-migrating components are included (Dai & Wang 1999). Pressure "knows about" the geometry of the whole atmosphere; temperature knows about the sun overhead (Figs. 2–3). Station-level daily P–T correlations range -0.05 to -0.56 with 95% CIs excluding any tight positive lock — variable in sign and magnitude, set by synoptic regime rather than by any gas-law coupling. **Band-resolved coherence** sharpens the bulk correlation (the blunt instrument) into the actual statement: magnitude-squared coherence between hourly P and T is 0.82–0.98 *at* the solar lines (S_1/S_2, where both fields are phase-locked to the same astronomical forcing; above the phase-randomized-surrogate 95% level at all eight stations) and collapses to the surrogate noise floor (0.09–0.30 vs floor 0.15–0.20) in the off-line sub-diurnal band (9–11 h and 13–20 h): away from the shared solar clock, nothing locks the two fields.

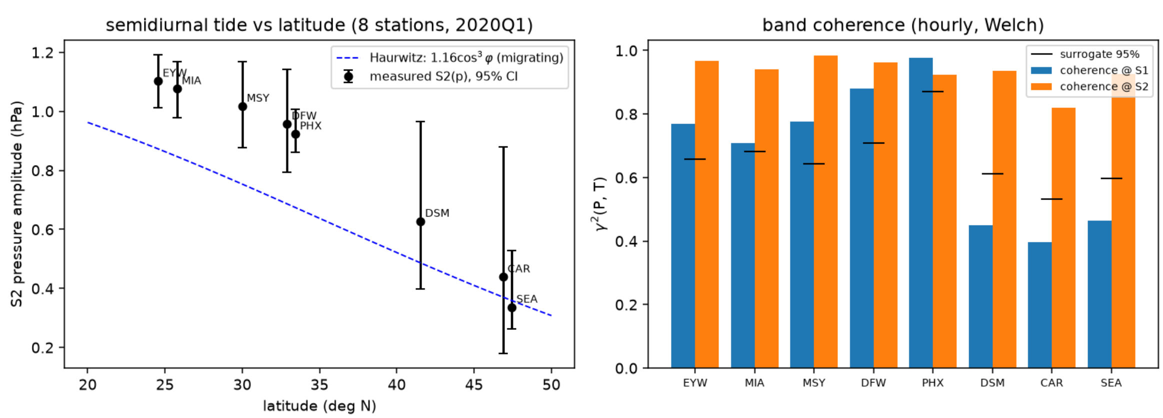

Figure 2. Left: semidiurnal pressure-tide amplitude vs latitude at eight stations with bootstrap 95% CIs, against the Haurwitz $1.16\cos^3\varphi$ migrating-tide climatology: monotonic equator-to-pole decay — a global resonance, not local heating. Right: band coherence — high at the shared solar lines, at the surrogate floor off-line.

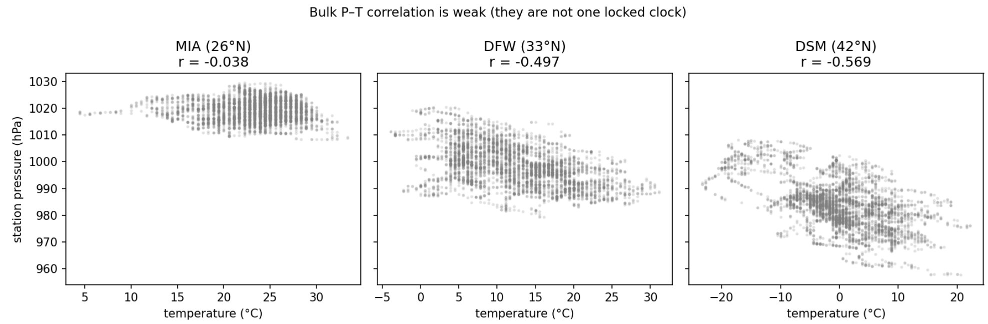

Figure 3. Daily-scale pressure–temperature coupling at the three ASOS stations: weak and regime-dependent, nowhere near a constant-density gas-law line.

Scope. Two months at the seasonal extremes (NEON) and one season (ASOS) at nine sites in one country: these are *fingerprint* measurements — the diurnal/semidiurnal split and its latitude ordering are the tested claims — not a global climatology. The tide amplitudes agree with the published climatological maps (Dai & Wang 1999) where they overlap.

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4. SPATIAL DECOUPLING MEASURED ON ONE GRID

4.1 Rayleigh–Bénard DNS: same forcing, two geometries

Field sensors cannot separate spatial structure; a DNS can. We use the 2-D Rayleigh–Bénard configuration of The Well (Ohana et al. 2024): $Ra = 10^6$, $Pr = 1$, 512×128, periodic in x, no-slip walls, with velocity, buoyancy, and pressure co-resolved on one grid. The decoupling is *not* about dominant sizes — both fields share the convection-roll wavelength (integral lengths 0.195 vs 0.207, ratio 1.06×). It lives at small scales (Figs. 4–5):

metric	buoyancy θ	pressure p	ratio p/θ
integral length (large scale)	0.195	0.207	1.06×
Taylor microscale (small scale)	0.253	0.607	**2.40**×
high-k spectral power ($k \approx 10$)	—	—	θ larger by $\sim \mathbf{10^4}\times$

Both numbers are the direct footprint of §2: $\hat{p} \propto \hat{f}/k^2$ suppresses spectral power by $\sim 1/k^4$, so pressure is smooth and blobby (elliptic) while buoyancy is advected into thin filaments with intense small-scale gradients (parabolic). Nothing was tuned: the same coupled simulation produces both fields.

Robustness across Ra. The single-snapshot 2.40× is not a one-Ra accident. Sweeping The Well's $Pr = 1$ test files at $Ra = 10^6, 10^7, 10^8$ (5 trajectories × 6 statistically-steady snapshots each) gives Taylor-microscale ratios of 2.58 [1.98–3.59], 2.94 [2.54–3.65], and 4.45 [3.27–5.88] (median [5–95%]; the headline snapshot sits inside the $Ra = 10^6$ interval), while the integral-length ratio stays order one (1.0–1.8) and the high-k spectral gap widens by another $\sim 70\times$. As Ra grows and buoyancy plumes sharpen toward finer scales, the inverse Laplacian keeps holding pressure at the large scales: *the elliptic–parabolic split widens with turbulence intensity rather than closing* (artifact `15b_rb_ra_sweep.json`).

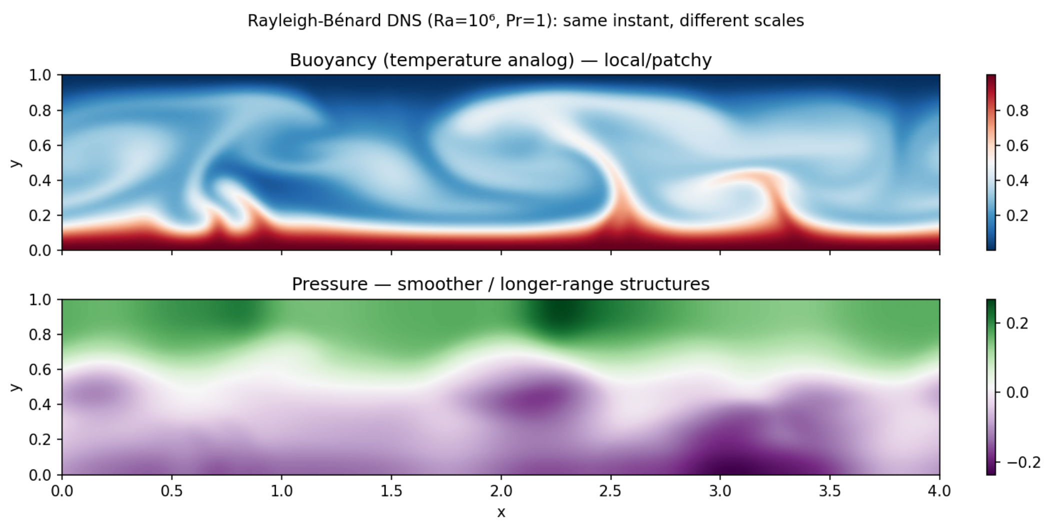

Figure 4. RB DNS snapshot: buoyancy (filamentary, sharp) vs pressure (smooth, domain-spanning) on the same grid at the same instant.

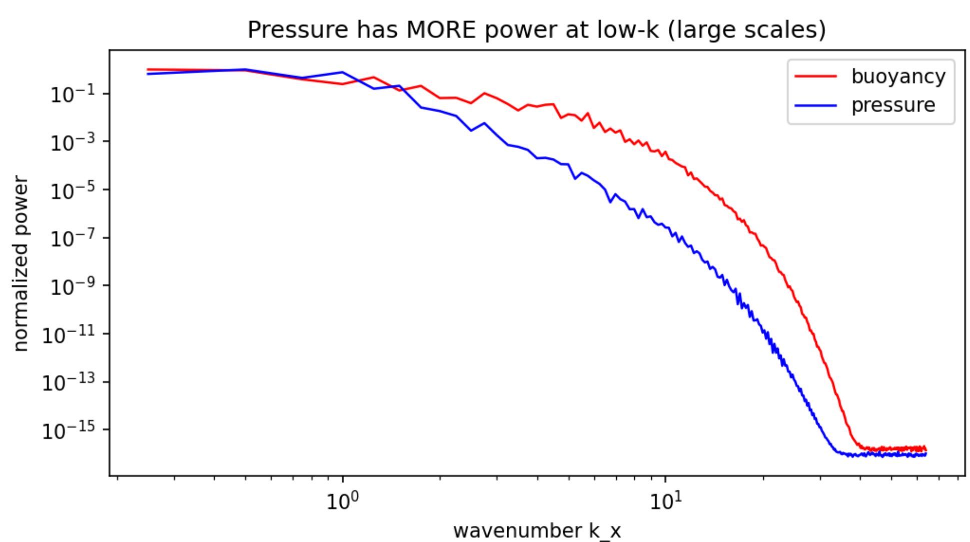

Figure 5. Spatial spectra: shared energy-containing scale; $\sim 10^4\times$ high-k gap — the $1/k^4$ elliptic suppression.

4.2 A single localized source: Green's functions in action

To isolate the operator contrast from turbulence statistics, we add a single localized bump to the bed of a developed cavity flow (frozen field), compute the continuity source it induces, $q = -\nabla\cdot(\text{bump}\cdot u)$, and apply that *same* source to both operators. The elliptic response $\Delta p = \nabla^{-2} q$ spreads over the whole cavity — **79%** of its magnitude lies beyond one bump-width of the source — while the heat-kernel (parabolic) response at matched time remains a confined Gaussian blob. Boundary geometry edits the *global* pressure field instantly, and only the *local* temperature field: any closure that models the pressure-borne flux with a local diffusivity is using the wrong Green's function.

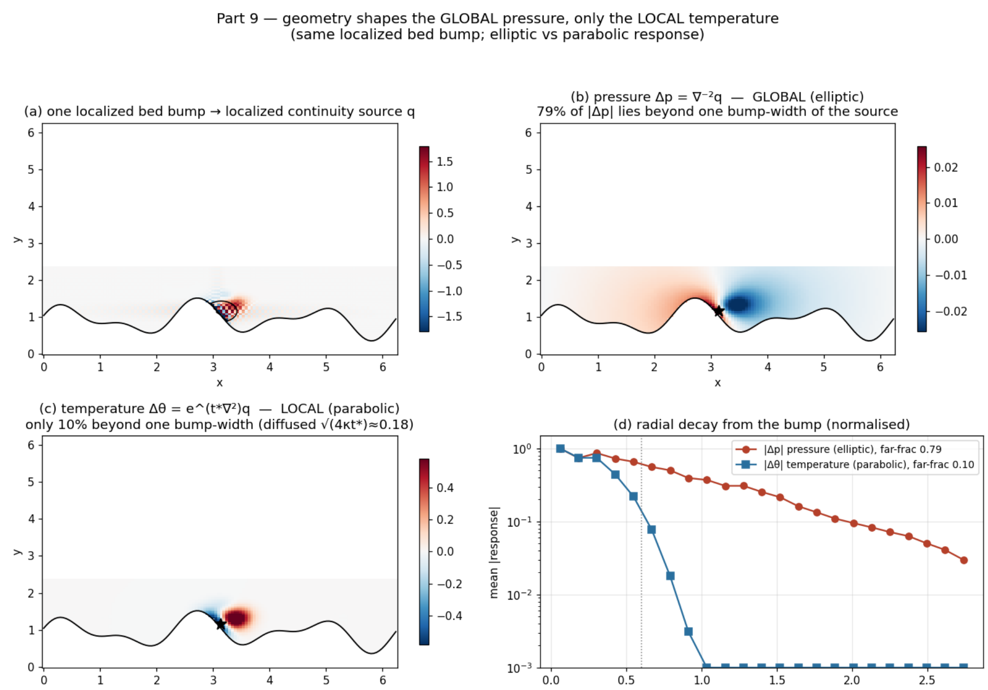

Figure 6. Identical localized source, two operators: global $\ln r$ elliptic response (79% beyond one bump-width) vs confined Gaussian parabolic response.

5. THE CLOSURE-RELEVANT CONSEQUENCE, MEASURED

If momentum is shaped by the global elliptic clock and heat by the local parabolic clock, then the *efficiencies* with which turbulence transports the two cannot stay in a fixed ratio as the shear/buoyancy balance changes. First-order K-theory with a constant turbulent Prandtl number ($K_M \approx Pr_t^{-1} K_H$) says they must.

Using the same NEON tower (602 usable 30-min periods), we bin the transport-efficiency correlations r_{uw} (momentum) and $|r_{wT}|$ (heat) by Monin–Obukhov stability $\zeta = (z - d)/L$ (Monin & Obukhov 1954; Foken 2008):

stability class	n	median r_{uw}	median $\lvert r_{wT}\rvert$	momentum/heat ratio
strongly unstable	95	0.182	0.209	0.87
unstable	73	0.324	0.150	2.17
near-neutral	121	0.312	0.095	**3.28**
stable	153	0.225	0.200	1.13
strongly stable	160	0.156	0.203	**0.77**

Momentum efficiency peaks near-neutral (shear-driven, pressure-mediated); heat efficiency peaks under strong convection (buoyant plumes) — they move *oppositely*, and their ratio swings **4.3**× (0.77 → 3.28). The summer month repeats the swing with more convective sampling: 0.43 → 2.60 across the same ζ classes (6.0×; 1030 usable periods), so the blind spot is not a winter artifact. The stability dependence itself is classical — turbulent-Prandtl-number variation across ζ is embedded in the Businger–Dyer flux–profile relations (Businger et al. 1971) and reviewed at length by Li (2019) — so the row-level numbers validate the protocol against known surface-layer physics; the claim we *use* is only the structural one that follows: A single fixed-Pr_t diffusivity is structurally unable to reproduce a 4.3× swing in this ratio; stability-dependent similarity functions absorb it empirically, which is precisely the point — the physics forces the closure to grow a second, stability-indexed degree of freedom because one clock is not enough. This is E3, and it is the field-data anchor for the operator audit in P1.

6. CONTROLLED DEMONSTRATIONS OF THE MECHANISM

6.1 The hyperbolic → elliptic crossover (linear)

A minimal 2-D linear-acoustics solver makes the elliptic-vs-hyperbolic statement concrete. An initial pressure bump radiates as a ring at speed c and decays by geometric spreading (the wave regime). Under a fixed forcing, the compressible pressure relaxes to the *same* field as the elliptic Poisson solve — final relative error $\sim 2 \times 10^{-3}$ for every c tested — and the approach time scales as

$$t_{90} \approx 2.3\, L/c, \tag{5}$$

with error histories for all sound speeds collapsing onto a single curve in the rescaled time ct/L (Fig. 7). c's only role is to set the clock; as $c \to \infty$ the adjustment becomes instantaneous and (3) degenerates into (2). This is why water ($c \approx 1500$ m/s) is "more fiercely elliptic" than air ($c \approx 340$ m/s).

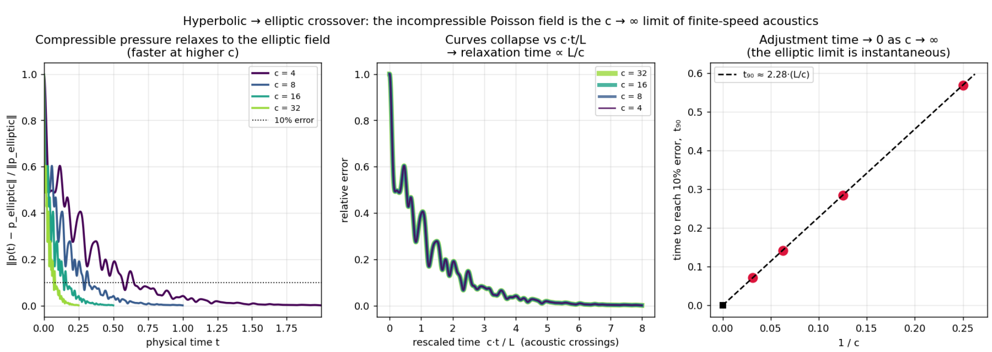

Figure 7. Crossover: relaxation of the compressible pressure to the elliptic field; all c collapse in ct/L; $t_{90} \approx 2.3\, L/c$.

6.2 Both clocks in one nonlinear flow

A fully nonlinear 2-D isothermal compressible Navier–Stokes solver (pseudo-spectral, periodic, $p = c^2\rho$ so c is an explicit knob) is initialized from a Taylor–Green vortex at $Re \approx 600$ and Helmholtz-split into solenoidal (vortical, slow) and dilatational (acoustic, fast) parts:

Mach M	$\langle KE_{dil}\rangle/\langle KE_{sol}\rangle$	avg-pressure residual vs elliptic
0.05	1.5×10^{-4}	8.1×10^{-4}
0.10	5.9×10^{-4}	3.2×10^{-3}
0.20	2.3×10^{-3}	1.4×10^{-2}
0.40	9.3×10^{-3}	1.0×10^{-1}

Two clocks coexist in one flow: the vortical energy decays on the slow advective time while the acoustic energy oscillates on the fast period (Fig. 8), and the fast clock's energy fraction scales as $\sim M^2$, vanishing in the incompressible limit (Fig. 9). The *instantaneous* pressure is dominated by the standing acoustic wave; averaging over the fast clock (the quasi-steady second half of each run) scrubs τ_p and leaves a field matching the elliptic Poisson pressure at correlation ≈ 1, with the residual itself shrinking as M^2. Averaging over the fast clock is a *low-pass filter that recovers the elliptic limit* — an operation §7 will repeat on the real atmosphere.

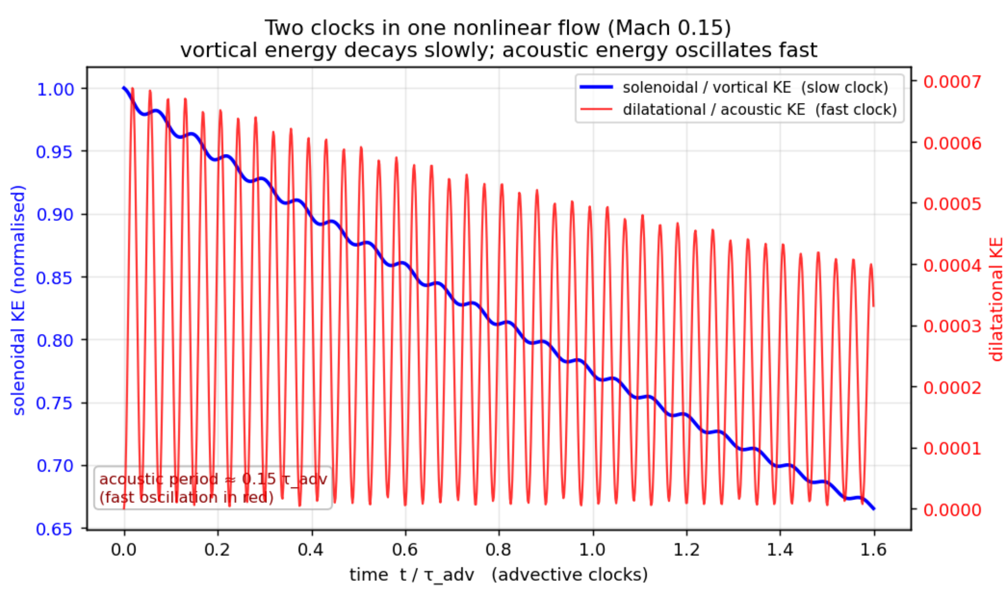

Figure 8. Nonlinear compressible box: slow vortical decay with fast acoustic oscillation riding on it — both clocks, one flow.

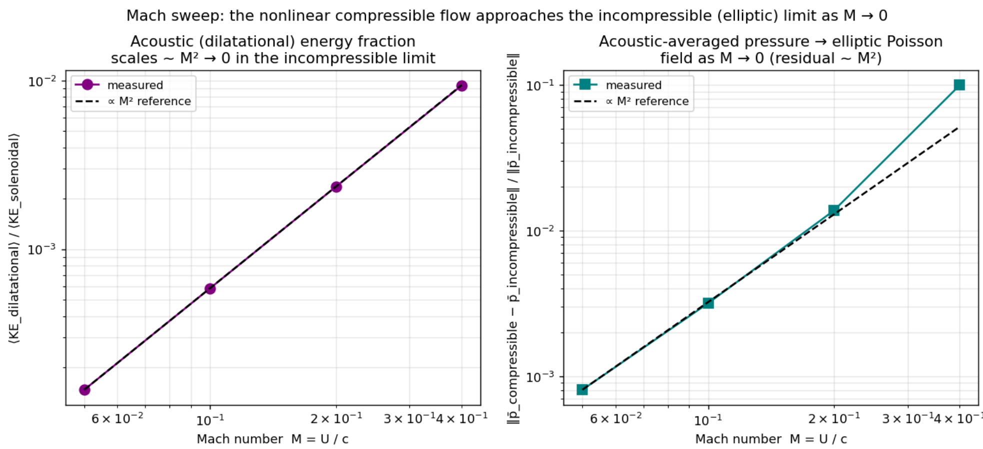

Figure 9. Mach sweep: dilatational energy fraction and acoustic-averaged pressure residual both $\sim M^2$.

6.3 The synthesis: projection is the two-clocks structure — and spectra cannot fake it

The standard projection (fractional-step) method advances a Boussinesq flow by a slow drift followed by an elliptic correction:

$$u^* = u^n + \Delta t\left[-(u\cdot\nabla)u + \nu\nabla^2 u + (b - \langle b\rangle)\hat{y}\right], \qquad u^{n+1} = \mathbb{P}u^*, \qquad b^{n+1} = b^n + \Delta t\left[-(u\cdot\nabla)b + \kappa\nabla^2 b\right], \tag{6}$$

with $\mathbb{P} = I - \nabla(\nabla^2)^{-1}\nabla\cdot$ the Leray projector (Leray 1934; Chorin 1968). Read structurally: the drift is the generator of the local Markov/heat semigroup — the *perfect* local, memoryless forecaster, exactly what single-K closure and statistical predictors represent — and the projection is one global Poisson solve per step, the fast clock acting as a constraint. Three results (Figs. 10–11):

- **The convection cell reappears from first principles**: plumes rise under the slow clock, the elliptic clock pulls in the compensating inflow (entrainment), and the shear between them rolls into vortices.
- **The local drift is divergent; one elliptic solve fixes it instantly.** The intermediate u^* violates mass conservation at RMS $\nabla\cdot u^* \approx 2.7\times10^{-2}$; a single projection drops it to $\approx 2\times10^{-15}$ — machine zero — through the global potential ϕ with $\nabla^2\phi = \nabla\cdot u^*$.

- **Energy yes, structure no.** Replacing the projection correction by a phase-randomized field with the *identical* power spectrum (Parseval-exact) — the structural move of spectrum-matched stochastic/SPDE closures (Hasselmann 1976; Berner et al. 2017) — yields pointwise correlation with the true correction of median $|r| = 0.07$ (95th percentile 0.20 over a 100-surrogate ensemble; the correction's energy sits in a few large-scale modes, so single-draw correlations scatter — we report the ensemble, not a favourable draw) and leaves RMS $\nabla \cdot u = 3.7 \times 10^{-2}$ [3.5–3.8×10^{-2}, 95%] instead of zero — no better than applying no correction at all (2.7×10^{-2}). Matching a spectrum does not reconstruct a boundary-aware, globally-coupled elliptic operator.

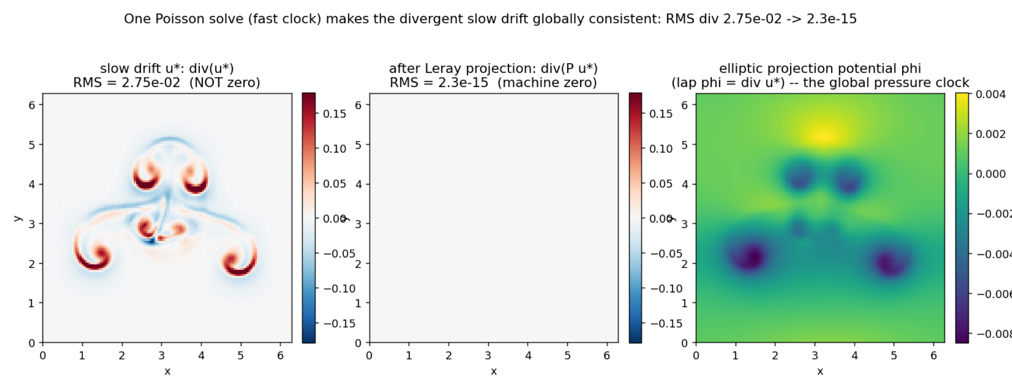

Figure 10. One time step, dissected: divergent local drift u^*; the global elliptic potential; the projected, solenoidal update (RMS $\nabla \cdot u$: $2.7 \times 10^{-2} \to 2 \times 10^{-15}$).

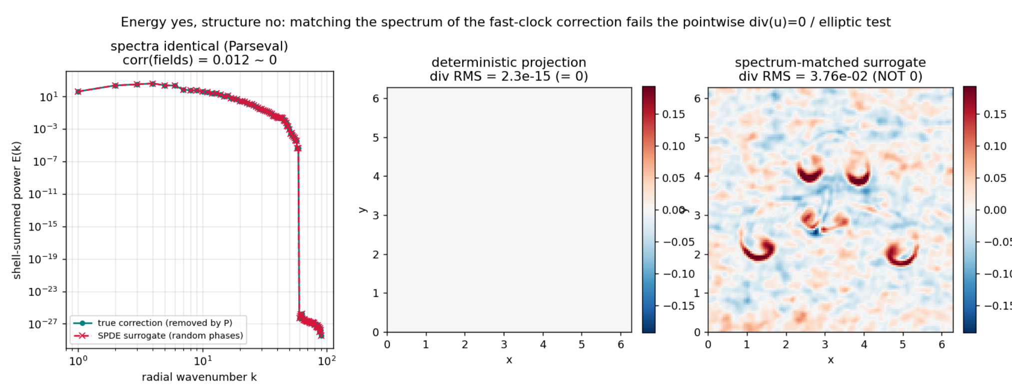

Figure 11. The SPDE ceiling: a Parseval-exact phase-randomized surrogate of the projection correction — correct spectrum, correlation median $|r| = 0.07$ (p95 0.20, $n = 100$), constraint unrepaired ($\nabla \cdot u \approx 3.7 \times 10^{-2}$, 95% interval 3.5–3.8×10^{-2}).

This is E5, and it is the sharpest structural statement in the paper: the fast clock is not "unresolved variance" to be re-injected with the right energy — it is a global operator with a definite geometric job. Stochastic closures that match spectra can repair energetics; they cannot repair geometry. (P1 builds the repair that can: a *projected*, dissipation-tied stochastic force — the projection is what rescues the noise.)

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7. THE TWO CLOCKS IN THE REAL ATMOSPHERIC WIND

Finally, the same decomposition on real winds. NCEP/NCAR Reanalysis 1 horizontal winds (2.5°; Kalnay et al. 1996), fetched live over OPeNDAP, are Helmholtz-decomposed on the sphere via the spherical-harmonic inversions $\nabla^2\psi = \zeta$, $\nabla^2\chi = \delta$ into a rotational (non-divergent, balanced — slow) and a divergent (fast) wind.

Honesty note. Horizontal divergence is not a model artifact: by 3-D continuity it is the genuine footprint of vertical motion. The diagnostic is therefore the energy split, not the existence of divergence.

level	R1: KE_{div}/KE_{rot} (daily mean)	R2 cross-check
850 hPa	4.0%	4.5%
500 hPa	**0.7%**	**0.8%**
250 hPa	1.0%	1.1%

- **The weather is rotational; the fast clock is a weak, structured residual.** The rotational wind *is* the synoptic highs/lows and jets; the divergent wind is $\sim$100× weaker in energy and concentrated in convergence/overturning zones — tropical convection, storm inflow — the real-atmosphere analogue of the dilatational component (Fig. 12).

- **Divergent $\ll$ rotational at every scale.** Splitting kinetic energy by spherical wavenumber l, the divergent branch sits 1–2 decades below the rotational branch for all resolved l; integrated, $KE_{div}/KE_{rot} \approx 0.04$ at 850 hPa — the real-data echo of $KE_{dil}/KE_{sol} \sim M^2 \ll 1$ (Fig. 13).

- **The vertical structure is physics, not tuning.** The divergent fraction is smallest near 500 hPa — the classical *level of non-divergence* (Holton 2004) — and larger in the boundary layer and jet-outflow levels (Fig. 14). **Cross-checked in a second assimilation system:** recomputing the same window with the same Helmholtz split from NCEP–DOE Reanalysis 2 (different model physics and assimilation corrections; Kanamitsu et al. 2002) gives 4.5%/0.8%/1.1% — same ordering, same 500-hPa minimum, ≤ 15% relative differences (`run_reanalysis_r2.py`, `figures/27b_reanalysis_crosscheck.json`); the ratio is a property of the analyzed atmosphere, not of one reanalysis.

- **Time-averaging is the low-pass filter.** The 6-hourly ratio at 500 hPa (1.1%) exceeds the daily-mean ratio (0.7%): averaging scrubs the fast divergent clock — the same operation that recovered the elliptic pressure from the acoustic field in §6.2.

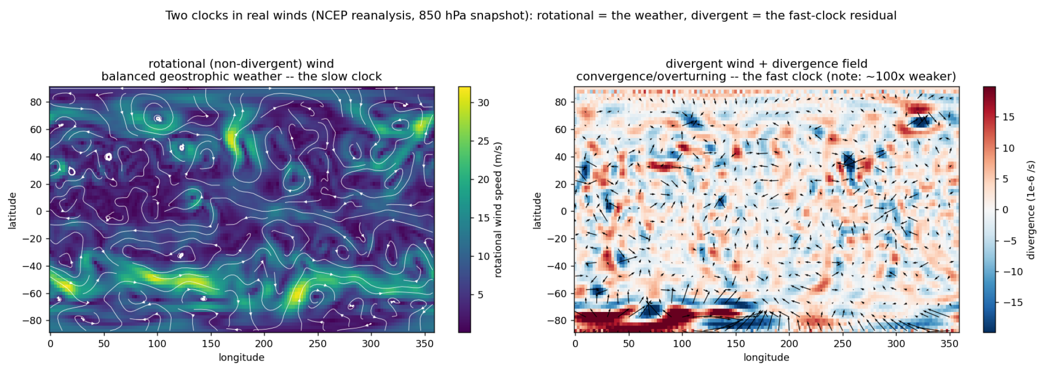

Figure 12. NCEP/NCAR winds, one analysis day: rotational wind (synoptic systems) vs divergent wind (weak, confined to overturning zones).

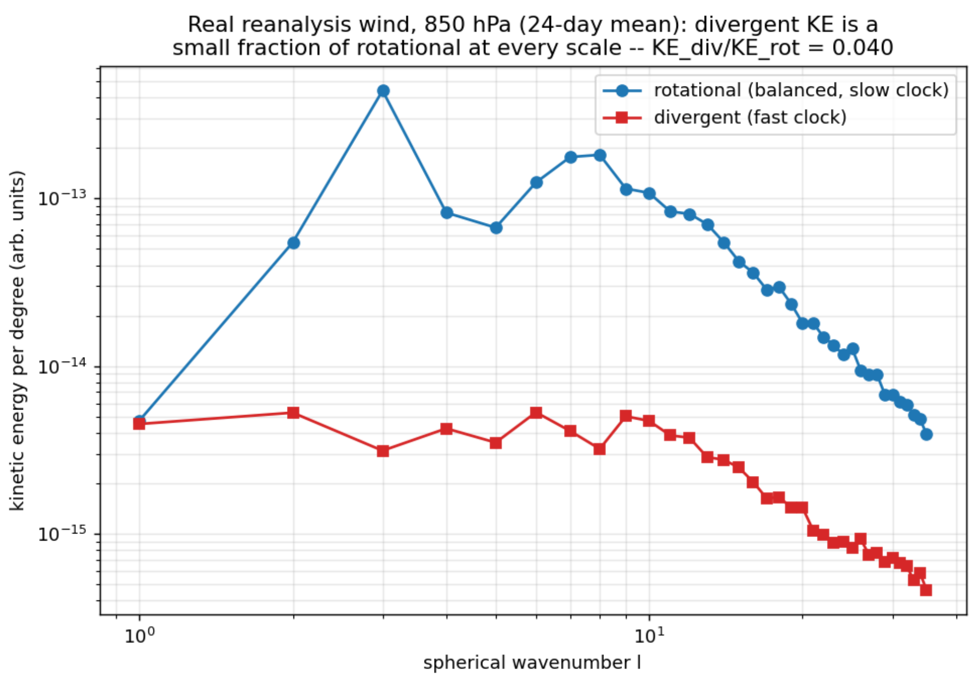

Figure 13. Kinetic-energy spectra by spherical wavenumber: the divergent branch 1–2 decades below the rotational at every $l \leq 35$.

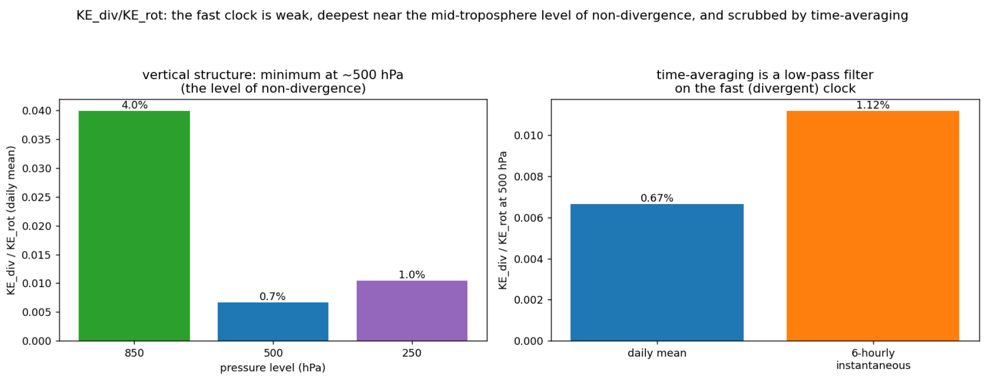

Figure 14. Vertical structure of KE_{div}/KE_{rot}: minimum at the level of non-divergence; 6-hourly > daily — time-averaging as low-pass.

Scope. Reanalysis is a model-assimilated product, not raw observation, and 2.5° resolves only $l \leq 35$. This section is a real-data *confirmation* of the energy split and its structure — not a turbulence-closure claim and not a gravity-wave climatology.

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8. DISCUSSION

8.1 What has been established

Six independent measurements, one structure. At a fixed site pressure and temperature run on different clocks with regime-dependent, never gas-law-locked bulk daily coupling (E1); on one DNS grid pressure lacks exactly the small scales the inverse Laplacian says it must lack (E2); the momentum/heat transport-efficiency ratio swings 4.3× in winter and 6.0× in summer where one-clock closure requires a constant (E3); finite compressibility turns the constraint into waves whose only slow-limit role is the clock $t_{90} \approx 2.3L/c$, with fast-clock energy $\sim M^2$ (E4); the fast clock's action on the slow dynamics is a global elliptic correction that spectrum-matched noise cannot imitate — correlation 0.01, constraint unrepaired (E5); and the real atmospheric wind carries $\lesssim 4\%$ of its energy in the fast clock, structured exactly where the physics puts it (E6).

8.2 Kill conditions (what would have falsified, or still would)

claim	observation that kills it	status
E1	tight positive daily P–T lock at fixed sites; tide amplitude ordered by local heating rather than latitude	not observed ($r \approx +0.07$ winter, -0.49 summer; tide 1.08 $\to$ 0.63 hPa poleward)
E2	pressure microscale $\approx$ scalar microscale; no high-k suppression	not observed (2.40×; 10^4×)
E3	transport-efficiency ratio constant (±tens of %) across stability	not observed (4.3× swing)
E4	relaxation time not $\propto L/c$; compressible field $\not\to$ Poisson field	not observed (collapse in ct/L)
E5	spectrum-matched surrogate reproducing the projection correction pointwise	not observed (ensemble median $\lvert r\rvert = 0.07$, p95 0.20; div unrepaired)
E6	$KE_{div} \sim KE_{rot}$, or structureless divergent wind	not observed ($\leq$ 4%, structured)

Each claim remains open to stronger tests: more sites and seasons (E1), higher-Ra and 3-D DNS (E2), other towers and surface types (E3), 3-D compressible sweeps (E4), wall-bounded domains where projection and filtering no longer commute (E5; see P1 §5d), and higher-resolution reanalyses or raw soundings (E6).

8.3 What this means for closures — and what it does not

The measurements above do not say K-theory is useless; they say it is *one-clock* physics, calibrated where the slow clock dominates and structurally silent about the fast one. Concretely: (i) any single-diffusivity model inherits E3's contradiction as soon as its calibration regime changes; (ii) any stochastic augmentation that matches spectra but not the elliptic geometry inherits E5's ceiling; (iii) any scheme that time-averages its inputs has already low-passed the fast clock and should not expect to recover fast-clock phenomena downstream (E4/E6). The constructive direction — collapse of the Mori–Zwanzig memory kernel, restoration of the projected fluctuation–dissipation force, the resulting benchmark closures in 2-D and 3-D — is the companion paper's subject (P1; cf. Chorin, Hald & Kupferman 2000; Parish & Duraisamy 2017). This paper deliberately claims none of it.

Equally deliberately, nothing here is a regularity statement (no Beale–Kato–Majda claim; the demonstrations are 2-D and periodic), nothing is an operational forecast-skill claim, and the reanalysis section stands on an assimilated product at coarse resolution. The claims are exactly the six numbers of §8.1, at the stated scopes.

9. CONCLUSION

Pressure and temperature share a source and a fluid but not a clock: one is a global elliptic/acoustic adjustment, the other a local parabolic transport. That split is measurable in a forest, at three airports, in a convection DNS, in two compressible solvers, in a projection step, and in the planetary wind field — six falsifiable expectations, six confirmations, with quantitative kill conditions on record. Its practical bite is structural: one-diffusivity closures and spectrum-matched stochastic augmentations are blind to the fast clock's geometry, measurably so (4.3× efficiency-ratio swing; ensemble-median correlation 0.07 of spectrum-matched surrogates with the true elliptic correction, divergence left unrepaired). The phenomenon established here is the load-bearing premise of the closure audit and repair developed in the companion paper.

APPENDIX A — REPRODUCIBILITY

All figures and numbers regenerate from the public repository (single commands, fixed seeds, CPU-only for every result in this paper):

result	command	figures
§3.1 NEON two clocks	`python general_two_clocks/run_analysis.py`	1 (and lead–lag, spectra)
§3.1 seasonal cross-check	`python general_two_clocks/run_neon_compare.py`	— (artifact)
§3.2 ASOS latitude sweep + CIs	`python general_two_clocks/asos/fetch_1min.py && python general_two_clocks/run_asos_ci.py`	15–20
§4.1 RB DNS split	`python general_two_clocks/run_rb.py`	4–5
§4.1 Ra sweep (10^6–10^8)	`python general_two_clocks/run_rb_ra_sweep.py`	— (artifact)
§4.2 Green's-function demo	`python general_two_clocks/run_elliptic_demo.py`	6
§5 stability bins	`python general_two_clocks/run_stability.py`	— (table)
§6.1 linear crossover	`python general_two_clocks/run_compressible.py`	7
§6.2 nonlinear Mach sweep	`python general_two_clocks/run_compressible_ns.py`	8–9
§6.3 projection/SPDE	`python general_two_clocks/run_boussinesq.py`	10–11
§7 reanalysis split	`python general_two_clocks/run_reanalysis.py`	12–14

Data endpoints: NEON DP4.00200.001 (data.neonscience.org, public); ASOS 1-minute via the Iowa Environmental Mesonet archive (mesonet.agron.iastate.edu, public); The Well RB dataset (polymathic-ai, public); NCEP/NCAR Reanalysis 1 via NOAA PSL OPeNDAP (psl.noaa.gov, public, no credentials). The repository's test suite (656 passing tests at draft time) covers the analysis modules; each report in `general_two_clocks/` documents per-part method details and robustness checks.

DATA AVAILABILITY

All analysis code, committed figure data, and the manuscript source are in the public repository at https://github.com/abdh0saifelden2-debug/K. All input datasets are public as listed in Appendix A.

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